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An OCR-Based Framework for Automated Verification of Thai and English Academic Transcripts

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***Abstract*—Academic transcripts are essential documents for employment and higher education but remain susceptible to forgery and manipulation, creating a need for efficient and reliable verification methods. Traditional verification is often time-consuming due to the structural complexity of transcripts and the challenges of extracting information from PDF and image formats. This study presents an OCR-based framework for automated transcript verification that integrates preprocessing and postprocessing techniques to enhance data extraction quality. English transcripts were evaluated using OCR models PaddleOCR, Tesseract, and EasyOCR, while Thai transcripts were assessed with Tesseract and EasyOCR. Experimental results demonstrate that preprocessing substantially improves extraction accuracy and that postprocessing further refines the outputs. Among the evaluated models, PaddleOCR achieved the highest performance on English transcripts with an overall accuracy of 81.87%, whereas Tesseract yielded the best accuracy for Thai transcripts at 81.11%. These findings underscore the effectiveness of combining OCR with tailored preprocessing and postprocessing strategies to support reliable and efficient transcript verification in academic settings.**

***Index Terms*—Optical character recognition (OCR), PaddleOCR, Tesseract, EasyOCR, Academic transcripts, Document classification.**

I. Introduction

In the current landscape of employment and higher education, applicants are typically required to submit supporting documents such as resumes, identification, certificates, and academic transcripts. Among these, transcripts play a critical role in validating academic achievements and qualifications. However, they are also vulnerable to alteration and forgery, creating a pressing need for reliable and efficient verification mechanisms.

Transcript verification at King Mongkut's Institute of Technology Ladkrabang (KMITL) is still conducted through conventional channels. External organizations—such as companies or universities—submit inquiries via email or postal mail to the registrar's office, where staff manually compare the submitted documents, usually in PDF or scanned form, against the official database. Although transcripts at KMITL follow a standardized layout, this approach remains slow, error-prone, and increasingly burdensome, with verification requests already exceeding 2,500 cases annually and continuing to rise.

Optical Character Recognition (OCR) technology provides a promising pathway to address these challenges by automatically extracting textual information from non-editable transcript files (e.g., PDFs or scanned images) and converting it into structured digital data. When combined with appropriate preprocessing to enhance image quality and postprocessing to refine textual outputs, OCR enables accurate alignment with institutional records, thereby accelerating verification and reducing the likelihood of human error. The adoption of OCR-based verification can thus transform transcript validation at KMITL into a faster, more accurate, and scalable system that strengthens institutional efficiency and builds trust with external stakeholders.

This study develops and evaluates an OCR-based framework specifically tailored for KMITL's transcript verification. The framework integrates preprocessing techniques, applies multiple OCR models for both Thai and English transcripts, and incorporates postprocessing methods to improve accuracy. Experimental evaluations were conducted on widely used OCR models, including PaddleOCR [1], Tesseract [2], and EasyOCR [3]. Results show that preprocessing and postprocessing substantially enhance extraction quality, with PaddleOCR achieving the highest accuracy for English transcripts and Tesseract performing best for Thai transcripts. Overall, this research not only delivers a practical solution for transcript verification at KMITL but also establishes a foundation for scalable applications in academic institutions, contributing toward secure, efficient, and technology-driven academic document verification in higher education institutions.

II. Related Works

Optical Character Recognition (OCR) has long served as a foundational technology for digitizing textual content. By converting printed or handwritten text in images or scanned documents into machine-readable formats, OCR enables efficient storage, retrieval, and analysis of paper- and image-based records. Modern OCR engines, enhanced by machine

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learning and deep learning, have demonstrated robust recognition capabilities across diverse fonts, layouts, and even handwritten styles. Beyond digitization, OCR has increasingly been integrated into higher-level applications such as information extraction, natural language processing (NLP), and document understanding.

Research has also focused on addressing language-specific challenges within OCR-based pipelines. For example, a notable study investigated Thai spelling correction from OCR word outputs [4]. The Thai language presents unique challenges due to its complex script system—44 consonants, 12 vowels, 4 tone marks, and 10 special characters—together with the absence of explicit word boundaries. These characteristics complicate downstream applications such as sentiment analysis, machine translation, and OCR postprocessing. The study introduced a two-stage framework of error detection and correction. Using 133,757 sentences from the Thai Digital Collection (TDC), containing over 224,000 erroneous syllables, synthetic text-image datasets were generated across multiple fonts and sizes to simulate OCR scenarios. Tesseract OCR was used to extract text, exposing common errors such as missing characters and improper line breaks. Experimental comparisons showed that Bi-LSTM achieved the best performance for error detection. These results underscore the role of advanced deep learning methods in enhancing Thai OCR postprocessing.

Beyond language-specific tasks, general-purpose OCR-based approaches have targeted structured information extraction. The FUNSD dataset [5], derived from RVL-CDIP, provides annotated forms with entities labeled as key, value, header, or other, along with semantic relationships. While the dataset proved useful, annotation inconsistencies limited its broader applicability. Subsequent work revised the dataset and introduced U-Net models enhanced with Channel-Invariant Deformable Convolutions for improved key-value pair detection. These efforts highlight how spatial layout analysis and structured annotations can augment OCR-based approaches for practical document understanding.

In contrast to OCR-dependent pipelines, recent studies have advanced OCR-free methods for document understanding. The Document Understanding Transformer (DONUT) [6] was introduced as an end-to-end framework that directly encodes document images into structured outputs such as JSON, thereby eliminating the need for OCR. By pretraining on large-scale datasets, including IIT-CDIP and the synthetic SynthDoG corpus, DONUT demonstrated superior accuracy, efficiency, and generalizability compared to OCR-based visual document understanding systems, marking a pivotal step toward scalable OCR-free solutions. Building on this line of work, an OCR-free approach for identity document information extraction [7] combined a Swin Transformer encoder with a BART decoder to jointly capture visual layouts and semantic content. To mitigate data scarcity, the study generated more than 7,000 synthetic documents in addition to 969 real Italian identity records. Experimental results showed that models pretrained with synthetic data substantially outperformed those trained only on real data, particularly in low-resource scenarios. While OCR-free methods have proven highly effective in structured domains such as identity documents, preliminary experiments in applying DONUT to educational transcripts suggest challenges in generalizing to the more complex and heterogeneous layouts of transcript data.

Taken together, these studies illustrate both the advances and challenges in OCR and document understanding. OCR-based pipelines remain indispensable for digitizing textual information and handling language-specific contexts such as Thai, but they often require sophisticated postprocessing to overcome recognition errors. OCR-free models such as DONUT demonstrate promising alternatives for structured document interpretation and identity verification, yet their adaptability to domain-specific applications like academic transcripts remains limited. This gap underscores the need for tailored OCR frameworks that integrate preprocessing, robust recognition, and postprocessing to enable reliable verification in transcript-focused scenarios.

III. Methodology

The proposed methodology for developing an OCR-based framework for the automated verification of Thai and English academic transcripts at KMITL is organized into five sequential stages, as illustrated in Figure 1. The workflow begins with image acquisition, where transcript images are collected for further processing. This is followed by data preprocessing, which enhances image quality and prepares the documents for recognition through techniques such as denoising, binarization, and deskewing. In the information extraction stage, OCR models are applied to detect and convert textual content from the transcript images into machine-readable text. The extracted text then proceeds to the postprocessing stage, where it is transformed into a structured key–value JSON format and further refined through error correction techniques leveraging domain-specific rules, dictionaries, and statistical models. Finally, the validated JSON output is cross-referenced with the KMITL Registration and Educational Service Office (KRESO) database, the authoritative source of student records, thereby enabling automated verification of transcript content and consistency.

A. Image Acquisition and Data Preprocessing

The process begins with the acquisition of transcript documents, which may be provided as scanned images, PDF exports, or captured images. In cases where the input is a PDF file, it is first converted into image format (.png), with a resolution set at 300 DPI to ensure optimal quality for subsequent processing.

Once converted, the images are passed through a Transcript Classification Model to verify whether the input corresponds to an academic transcript. The dataset used for developing this model comprised a total of 9,130 images, including 3,510 transcript document from King Mongkut's Institute of Technology Ladkrabang (KMITL) and 5,620 non-transcript images collected from various sources. These non-transcript samples

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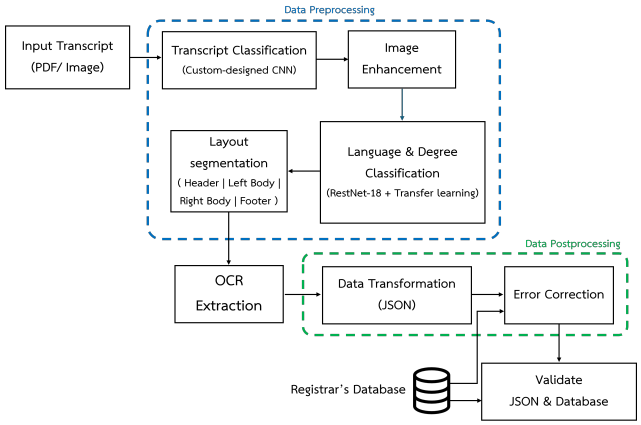


Fig. 1. End-to-end pipeline of the proposed OCR-based methodology for automated verification of Thai and English academic transcripts.

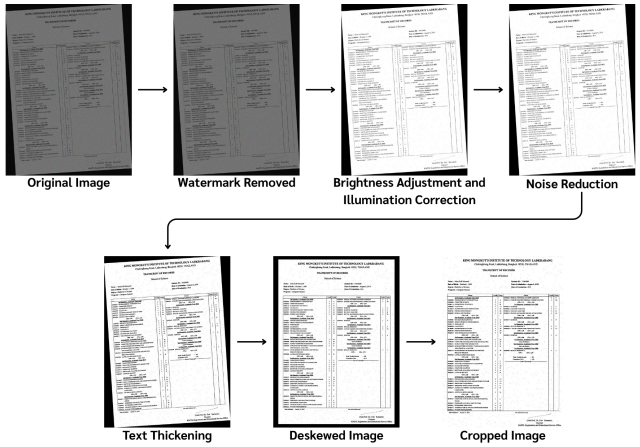


Fig. 2. Image Enhancement Workflow in Data Preprocessing.

included diverse document types such as textbooks (e.g., Physical Science Interactive Textbook, Gr10 Mathematics, Basic Mathematics, NCERT Class 10 Science), financial and receipt datasets, stamp verification datasets, scanned images, and even natural scene and pet image datasets. This diversity was intentionally included to ensure the model's robustness in distinguishing transcript documents from irrelevant or misleading inputs.

The dataset was divided into training (72%), validation (18%), and test (10%) sets. A custom deep learning model based on a Convolutional Neural Network (CNN) with a sequential architecture was employed, consisting of convolutional layers followed by max pooling, flattening, and fully connected dense layers with ReLU activation. The final classification layer used a softmax function for binary classification. The model was trained using the Adam optimizer and categorical cross-entropy as the loss function. The evaluation results demonstrated a high level of performance, achieving an accuracy of 99.89% with a loss of 1.41%. If the input is classified as non-transcript, the system immediately alerts the user and terminates the process. Conversely, if identified as a valid transcript, the image is forwarded to the image enhancement stage.

In this stage, several preprocessing operations are applied to improve the image quality for OCR. First, watermarks or colored institutional stamps are suppressed by filtering high-saturation and high-value pixels and replacing them with white, thereby enhancing text prominence. Next, brightness adjustment and illumination correction are performed, followed by noise reduction to further enhance the continuity of text strokes. Since some fine details may be lost, a character thickening process is applied to restore clarity and improve OCR recognition. Deskewing is then carried out by detecting skew angles via the Hough Line Transform and rotating the image accordingly. Finally, the document region is cropped to exclude unnecessary background elements. The results of each enhancement step are illustrated in Figure 2. After enhancement, image quality is reassessed to ensure the document remains valid as a transcript. If the quality is insufficient, the user is notified to recheck the input; otherwise, the document proceeds to the Transcript Language and Degree Classification Model. This model determines the language (Thai or English) and the academic level (undergraduate or graduate). It was trained on 3,510 transcript images evenly distributed across the four target classes, with additional non-transcript samples included for robustness. The dataset was divided into training (80%) and testing (20%) subsets. A ResNet-18 architecture with pretrained weights was adopted as the feature extractor, and transfer learning was applied for multi-output classification. The model demonstrated strong generalization ability on the test set, achieving highly satisfactory results for both tasks.

Following classification, the system performs table structure analysis to segment each transcript into four logical regions: header, left body, right body, and footer. Morphological operations with horizontal and vertical kernels are applied to detect table lines, followed by region-based cropping. To address misalignment, the widths of the left and right body regions are compared, and if a predefined threshold is exceeded, the middle block is adjusted and re-divided. Aspect ratios of the cropped regions are also validated to ensure structural consistency. Validated segments are then forwarded for subsequent processing, while failures trigger an error message indicating segmentation issues.

B. Information Extraction

After segmentation, each region enters the Information Extraction stage, where Optical Character Recognition (OCR) is applied to convert visual characters into machine-readable text. To improve reliability, postprocessing techniques such as dictionary-based spell-checking are employed. In this study, three widely adopted OCR frameworks were evaluated: EasyOCR, PaddleOCR, and Tesseract. EasyOCR supports more than 80 languages, including Thai and English, making it suitable for multilingual tasks with minimal configuration. PaddleOCR, developed under the PaddlePaddle deep learning platform, provides high-performance recognition across

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multiple languages but does not include official support for Thai. Tesseract, one of the most established open-source OCR engines, supports over 100 languages including Thai and English, and since version 4.0 integrates Long Short-Term Memory (LSTM) networks to improve recognition, particularly for complex scripts and degraded images. By benchmarking these frameworks on both Thai and English transcripts, the system ensures accurate and consistent text extraction tailored to the requirements of KMITL's verification process.

C. Data Transformation and Error-correction

The outputs generated by OCR models are inherently heterogeneous and must undergo transformation to ensure consistency with the database schema. At this stage, error-correction mechanisms are also integrated. The extracted text is reformatted into a structured key–value representation, enabling direct comparison with reference records stored in the KRESO database. Specifically, PaddleOCR and EasyOCR produce structured lists containing bounding box coordinates, recognized text, and confidence scores, whereas Tesseract generates plain text without explicit structural annotations. Consequently, the transformation pipeline is adapted to each framework.

The process begins with the Header, which serves as a primary quality indicator. If critical attributes (e.g., student name or ID) are missing, and if fewer than 70% of required header fields are successfully extracted, the document is flagged as low quality and rejected. Otherwise, the Body and Footer are transformed into the predefined schema. For PaddleOCR and EasyOCR, line merging is performed based on y-axis alignment, while for Tesseract segmentation is guided by predefined keywords (e.g., "Name," "Student ID," "Degree"). Preprocessing operations, including case normalization, whitespace trimming, and removal of extraneous symbols, are applied uniformly across all frameworks.

Error-correction is integrated throughout the document. In the Header, fuzzy matching based on Levenshtein distance is employed to mitigate OCR-induced variations, and similarity with reference keywords (e.g., institutional name) is accepted when exceeding 80%. Academic honors are normalized into numeric codes (e.g., 1 = first-class honors, 2 = second-class honors), and date formats are standardized by converting month names and synchronizing Gregorian and Buddhist calendar years.

In the Body, transcript content extracted from left and right pages is merged and organized by semester. Core attributes include academic year, semester, credit units, GPA, course code, course title, and grade (e.g., A–F, S/U, O, G, P, I, T(A–B)). Graduate transcripts has additionally specify course types (e.g., Cr, Nc, Tr), and special cases such as thesis defenses are identified through domain-specific keywords. Error-correction in this section emphasizes course code validation: if a course code matches an entry in the database and the corresponding course title is sufficiently similar, the system corrects the course title to align with the reference record. GPA values and course grades are not altered but reformatted into standardized patterns (e.g., X.XX), thereby facilitating reliable database-level verification while preserving the integrity of the original course records. For graduate-level transcripts, examination-related fields are further validated against the graduate assessment database.

The Footer includes the date of issue, the name of the authorized registrar, and the position of the signatory. These elements are extracted and structured into the schema, after which error-correction procedures cross-check registrar names and positions against authoritative records.

Finally, all extracted and corrected information from the Header, Body, and Footer is consolidated into a JSON structure, producing a standardized key–value representation that ensures compatibility with automated verification workflows.

IV. Evaluation

The dataset used in this study comprises synthetically generated academic transcripts provided by the KMITL Registration and Educational Services Office (KRESO). All documents were anonymized to preserve confidentiality while maintaining the structural and linguistic characteristics of authentic records. The corpus covers both undergraduate and graduate levels, consisting of 211 undergraduate transcripts in English, 211 undergraduate transcripts in Thai, 140 graduate transcripts in English, and 140 graduate transcripts in Thai, for a total of 702 documents. Each transcript follows the standard structure of an academic record, including header fields (e.g., student name, program, degree level, graduation status), body sections (e.g., course lists, credit units, grades), summary fields (e.g., GPA), and footer elements (e.g., date of issue, signatory name, and position). Representative examples of these transcript types are illustrated in Figure 3.

To enhance model robustness and account for potential variations in real-world scenarios, data augmentation was applied to the original set of transcripts. Four augmentation techniques were employed: (i) random rotation within a range of -10° to $+10^{\circ}$, (ii) noise injection using five noise types (Gaussian, Salt, Pepper, Salt-and-Pepper, and Speckle) with varying variances, (iii) brightness adjustment by scaling image intensity between 20% and 200% of the original, and (iv) a combined augmentation simultaneously applying rotation, noise, and brightness adjustment. In total, these procedures generated 2,808 augmented images. Combined with the original 702 transcripts, the final dataset comprised 3,510 samples. The details of the augmentation methods and sample distributions are summarized in Table I.

The original transcript documents, together with their augmented counterparts, were evaluated under four experimental configurations: (i) without image enhancement or error correction, (ii) with image enhancement only, (iii) with error correction only, and (iv) with both image enhancement and error correction. Across all configurations, three OCR models—Tesseract, PaddleOCR, and EasyOCR—were applied to English transcripts, while Tesseract and EasyOCR were used for Thai transcripts. System performance was evaluated using

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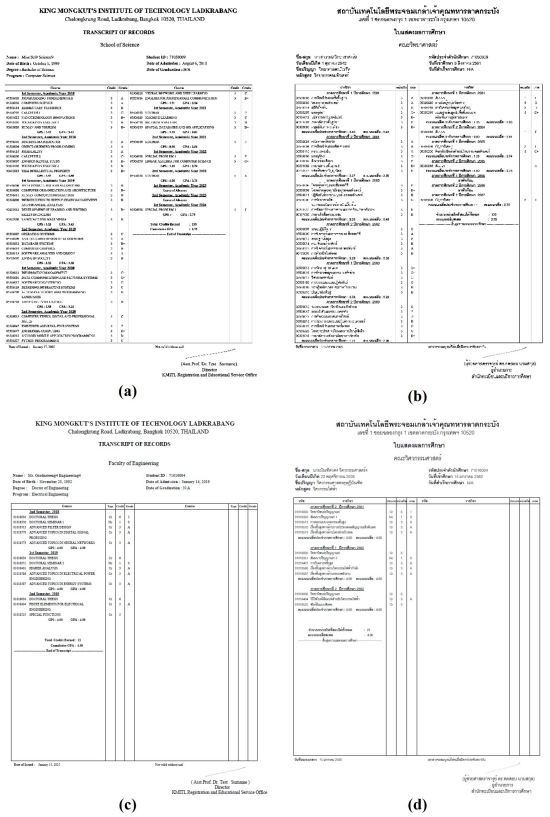


Fig. 3. Illustration of the original transcript documents used in this study, consisting of (a) undergraduate transcript in English, (b) undergraduate transcript in Thai, (c) graduate transcript in English, and (d) graduate transcript in Thai.

TABLE I
SUMMARY OF TRANSCRIPT DATASET INCLUDING ORIGINAL AND AUGMENTED SAMPLES

Dataset Category	Method	No. of Samples
Original transcripts	Undergraduate (English)	211
	Undergraduate (Thai)	211
	Graduate (English)	140
	Graduate (Thai)	140
	Total Original	**702**
Augmented transcripts	(i) Rotation ($-10°$ to $+10°$)	702
	(ii) Noise injection (5 types)	702
	(iii) Brightness adjustment (20%–200%)	702
	(iv) Combined augmentation	702
	Total Augmented	**2,808**
Grand Total		**3,510**

four metrics: Accuracy, F1-Score, Word Error Rate (WER), and Character Error Rate (CER). A summary of the results is presented in Table II.

In Configuration 1 (no enhancement and no correction), the raw transcript images were directly processed by the OCR models. Many documents exhibiting skew or rotation could not be segmented into the required four sub-sections, as the table lines used for segmentation were undetectable. Only documents with minimal skew (less than ±0.35°) and clearly defined grid lines were successfully processed. Nevertheless, documents with varying levels of noise or brightness were still segmentable and could undergo OCR. Among English transcripts, Tesseract achieved the best performance, followed by PaddleOCR and EasyOCR. For Thai transcripts, Tesseract again outperformed EasyOCR.

TABLE II
OCR MODEL PERFORMANCE ON ENGLISH (EN) AND THAI (TH) TRANSCRIPTS UNDER FOUR CONFIGURATIONS: RAW, ENHANCEMENT, CORRECTION, AND ENHANCEMENT + CORRECTION.

OCR Model	Doc. Type	Doc. Lang.	Accuracy	F1-Score	WER	CER
Tesseract	Raw	EN	0.5924	0.5926	0.5023	0.3673
	Enhancement Only		0.7575	0.7576	0.3855	.1987
	Correction Only		0.6091	0.6094	0.4743	0.3640
	Enh.+ Corr.		**0.7776**	**0.7776**	**0.3425**	**0.1964**
PaddleOCR	Raw	EN	0.5612	0.5620	0.5613	0.3454
	Enhancement Only		0.6740	0.6747	0.5089	0.1969
	Correction Only		0.6605	0.6612	0.4184	0.3121
	Enh.+Corr.		**0.8187**	**0.8193**	**0.2983**	**0.1486**
EasyOCR	Raw	EN	0.4904	0.4896	0.6343	0.4114
	Enhancement Only		0.5145	0.5093	0.6566	0.3364
	Correction Only		0.5803	0.5800	0.5048	0.3823
	Enh.+ Corr.		**0.6262**	**0.6212**	**0.4993**	**0.2968**
Tesseract	Raw	TH	0.4812	0.4817	0.6215	0.4082
	Enhancement Only		0.5973	0.5980	0.5712	0.2316
	Correction Only		0.6209	0.6220	0.4191	0.3790
	Enh.+ Corr.		**0.8111**	**0.8126**	**0.2497**	**0.1879**
EasyOCR	Raw	TH	0.4602	0.4616	0.6352	0.4353
	Enhancement Only		0.5423	0.5432	0.6236	0.2825
	Correction Only		0.5843	0.5849	0.4642	0.4035
	Enh.+ Corr.		**0.7164**	**0.7166**	**0.3784**	**0.2422**

In Configuration 2 (with enhancement but without correction), performance improved considerably. For English transcripts, Tesseract remained the top-performing model, followed by PaddleOCR and EasyOCR. For Thai transcripts, Tesseract again outperformed EasyOCR. Enhancement yielded substantial performance gains, particularly for Tesseract on English transcripts (+16.51%) and PaddleOCR on English transcripts (+11.28%). Similar improvements were also observed in Thai transcripts, where Tesseract improved by +11.61% and EasyOCR by +8.21%.

In Configuration 3 (without enhancement but with correction), extracted text was validated against mandatory attributes, including the student's name and student ID, and was required to contain at least 70% of the expected header information. This filtering ensured that only structurally valid outputs were processed further. Under this setting, PaddleOCR achieved the highest performance on English transcripts, followed by Tesseract and EasyOCR, whereas EasyOCR surpassed Tesseract on Thai transcripts. Post-processing applied directly to raw outputs provided notable improvements for Thai transcripts (Tesseract +13.97%, EasyOCR +12.41%) but only a minor gain for Tesseract on English transcripts (+1.67%). When correction was applied after enhancement, performance increased further, most prominently for Tesseract on Thai transcripts (+21.38%) and EasyOCR on Thai transcripts (+17.41%), as well as for PaddleOCR on English transcripts (+14.47%).

Finally, Configuration 4 (with both enhancement and correction) achieved the highest performance across all models and both languages. The combined approach consistently delivered the greatest improvements, with Tesseract on Thai

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transcripts improving by +32.99% and PaddleOCR on English transcripts by +25.75%, thereby highlighting the synergistic advantage of combining enhancement and correction strategies within the OCR pipeline.

Overall, the findings clearly demonstrate that enhancement substantially improves OCR robustness, particularly in addressing document skew and variations in image quality, while correction plays a critical role in ensuring both structural completeness and semantic validity of the extracted content. Among the evaluated OCR models, Tesseract consistently achieved the highest performance on English transcripts and also led in Thai transcripts, except under the correction-only setting, where EasyOCR exhibited superior adaptability. PaddleOCR, although not always the top performer, delivered competitive results, particularly in English transcript extraction. These outcomes highlight that effective dataset preparation, coupled with the integration of both enhancement and correction strategies, is essential for achieving reliable and accurate OCR performance on academic transcript documents.

V. Conclusion

This study presented an end-to-end pipeline for transcript digitization and verification that integrates image enhancement, OCR extraction, and correction for structured database validation. The framework begins with transcript classification, image enhancement, applies language- and degree-level classification, and then layout segmentation. The extracted text is subsequently transformed and corrected to ensure consistency and accuracy before being verified against the registrar's records.

Experimental evaluation on a synthetic dataset of 3,510 transcripts in English and Thai demonstrated the effectiveness of the proposed approach. Image enhancement substantially improved OCR robustness—most notably for English with gains up to +16.51% (Tesseract) and for Thai with gains up to +11.61% (Tesseract) over raw inputs. Applying correction further boosted reliability: when applied after enhancement, improvements reached +14.47% for English (PaddleOCR) and +21.38% for Thai (Tesseract). Combining both enhancement and correction yielded the highest overall performance, with aggregate gains up to +25.75% for English (PaddleOCR) and +32.99% for Thai (Tesseract) relative to raw inputs. Among the OCR models, PaddleOCR achieved the best English performance under the combined setting, while Tesseract consistently led on Thai, with EasyOCR showing competitive results in Thai—particularly under the correction-only condition.

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